

NIHAL POTHUNOORI

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EDUCATION

University of Illinois at Urbana-Champaign

Expected May 2027

B.S. Electrical Engineering - Urbana-Champaign, IL

Relevant Coursework: Analog Signal Processing; Semiconductor Electronics; Computer Systems & Programming

SKILLS

Programming: Python, C/C++, Matlab, Java, C#, .NET

Electrical: Analog & digital circuits, PCB/PCBA design (schematic + layout), I2C/SPI, op-amp signal conditioning, power regulation, embedded bring-up & debugging

Tools: Altium Designer, KiCad, LTSpice, Git, Arduino, Raspberry Pi, ROS2, OpenCV, TensorFlow/PyTorch; oscilloscope, multimeter, bench power supply

WORK EXPERIENCE

Undergraduate Researcher

Jan 2026 - Present

DASLABS, University of Illinois

Champaign, IL

- Implementing vision-language-action (VLA) models to identify stable rest/anchor locations for a soft robot during locomotion and manipulation tasks.
- Built a ROS2-based multimodal perception + evaluation pipeline in Python (RGB/RGB-D segmentation, logging, metrics) to benchmark rest-point selection reliability.

Web Development / IT Support

Aug 2025 – Jan 2026

College of Education, UIUC

Champaign, IL

- Built a .NET 8 C# library integrating with Azure Active Directory and automated permissions management for ~2,500 users; shipped web updates in HTML/C#.

Undergraduate Researcher

Aug 2025 – Jan 2026

Autonomous Materials Systems Group, University of Illinois

Champaign, IL

- Designed and built a robotic manipulator + gantry system for automated DCPD polymer sample preparation in a DARPA-funded lab project.
- Applied TensorFlow-based reinforcement learning for closed-loop process control; improved throughput by reducing sample prep time 75%.

Line Cook

June 2025 – August 2025

Toast Gastropub

Richmond, VA

LEADERSHIP

Project Lead

Aug - Dec 2025

SigRobotics, University of Illinois

Champaign, IL

- Designed a custom PCB/PCBA for a wearable silent-speech interface (MIT AlterEgo-inspired) capturing facial/neck neuromuscular signals during subvocalization and delivering bone-conduction audio feedback for discreet HCI.
- Led a 10-person team through schematic, layout, BOM, fabrication, assembly, and bring-up; performed soldering/rework and debugged analog front-end + MCU interfaces with oscilloscope/multimeter.

Hardware Lead

Jan - May 2025

iRobotics Micromouse, University of Illinois

Champaign, IL

- Led hardware design for maze-solving robots; ran weekly design reviews and taught PCB/PCBA fundamentals (15+ members).
- Fabricated custom PCBs in Altium with I2C/SPI sensor interfaces and analog filtering

PROJECTS

Laundry Robot

Jun 2024 - Jul 2024

- Built an autonomous mobile system to collect scattered clothing using an OpenCV pipeline and closed-loop control on a differential-drive platform.
- Designed trajectory planning + PID wheel control with encoder feedback in embedded C/C++; tuned parameters using logged telemetry.

Wearable EMG Instrument (2nd Place - International IEEE Arduino Contest)

Aug 2023

- Designed an EMG analog front-end (op-amps, band-pass filtering, ADC) and mapped muscle activation to musical notes.
- Prototyped/soldered circuitry and debugged noise/stability with oscilloscope and bench supply; iterated filter stages in LTSpice/Matlab.